

What I Learned From This Project

A personal reflection on the Prototyping Module — Arduino line-following robot

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When our team was handed a fixed kit of components and asked to build a line-following robot that could also avoid obstacles, I thought the hard part would be the wiring. It turned out the hardest and most rewarding part was everything the project taught me about how I think, how I work with others, and how I respond when things do not go the way I planned. This is what I want to carry with me and be able to speak about with confidence.

Patience is a skill, not a personality trait

Early on I assumed that once the code looked correct, the robot would behave. It did not. A command to turn left made the car steer right; motors refused to move below a certain power; a sensor that worked one minute failed the next. I learned that debugging is not about being frustrated faster — it is about staying calm and isolating one variable at a time. I stopped guessing and started testing deliberately, changing a single thing, observing, and writing down what happened. That habit of slowing down to speed up is something I now trust in myself, and it applies far beyond electronics.

Being wrong is part of getting it right

One of my clearest memories is being completely convinced about how the turning geometry worked — and being completely wrong. Reality corrected me, and I had to let go of my assumption to move forward. I used to see mistakes as something to hide. This project taught me to treat them as information. Every failed attempt — the half-circle turn, the box detour — narrowed the path toward the approach that finally worked. I now see iteration not as a sign that I started badly, but as the normal shape of real problem-solving.

Constraints made me more creative, not less

We were not allowed to add any hardware outside the kit. At first that felt limiting. In the end it pushed me to understand what I already had far more deeply — to tune, adjust, and squeeze results out of the exact components in front of me instead of wishing for better ones. I learned that resourcefulness often matters more than resources, and that a clear boundary can actually sharpen your thinking.

A team is stronger when the work is shared honestly

Working with Dodly, Ye-Htet, and Gabriel taught me how much smoother progress becomes when people communicate openly. We split the hardware, the code, the documentation, and the presentation, but the real value was in how we reported results back to each other — honestly, including the failures. I learned to explain my reasoning clearly, to listen when a teammate saw something I missed, and to trust others with parts of a system I could not watch over myself. Good collaboration is not about doing everything; it is about doing your part well and keeping the others informed.

I can explain technical work to people who were not there

Preparing the report and the presentations forced me to translate messy, hands-on work into something clear and structured. I learned to focus on the idea rather than reciting every specification, and to let the supporting material carry the details so I could speak naturally about the bigger picture. Being able to stand in front of others and tell the story of what we built — why it works and what we overcame — gave me real confidence in myself as a communicator.

What I am taking with me

This project was more than an assignment. It taught me persistence when a problem refuses to yield, humility when my assumptions are proven wrong, creativity inside tight limits, and the quiet satisfaction of watching something I helped build finally work. Most of all, it showed me that I can take an intimidating, unfamiliar challenge, break it into pieces, and see it through with a team. That is the confidence I will carry into whatever I do next — and the experience I will be proud to speak about for myself.

“The robot only moved forward once I stopped assuming and started listening — to the hardware, to my teammates, and to my own mistakes.”